SOFTWARE REQUIREMENTS SPECIFICATION

Web-Based ML Model Evaluator

Complete Requirements Table

Req ID	Type	Description	Priority	Acceptance Criteria (PASS/FAIL)	Rationale
FR-1.1	Functional	The system shall allow a new user to register using required account information.	High	PASS: A valid unique account is created. FAIL: Invalid input is accepted or the account is not created.	Enables access to the evaluator.
FR-1.2	Functional	The system shall authenticate registered users using valid login credentials.	High	PASS: Valid credentials establish a session and invalid credentials are rejected. FAIL: Authentication behaves otherwise.	Protects user accounts.
FR-1.3	Functional	The system shall allow an authenticated user to log out and terminate the active session.	Medium	PASS: The previous session cannot access protected resources after logout. FAIL: The session remains usable.	Prevents continued access after logout.
FR-1.4	Functional	The system shall restrict protected resources and operations to authenticated users with the required permissions.	High	PASS: Anonymous and unauthorized requests are denied. FAIL: Protected resources are accessible without permission.	Protects user-specific resources.
FR-2.1	Functional	The system shall allow an authenticated user to upload a dataset in CSV format.	High	PASS: A valid CSV is accepted and associated with its uploader. FAIL: Valid CSV is rejected or ownership is incorrect.	Provides input data for experiments.
FR-2.2	Functional	The system shall validate uploaded files for supported format, readability, and usable tabular structure.	High	PASS: Valid CSV is accepted; empty, malformed, or unsupported files are rejected with feedback. FAIL: Invalid input is accepted.	Prevents unusable input.
FR-2.3	Functional	The system shall display the selected dataset's column names and a sample of rows.	High	PASS: Preview matches the selected dataset. FAIL: Rows or column names are missing or incorrect.	Lets users inspect data.
FR-2.4	Functional	The system shall display basic dataset statistics, including row and column counts and available data-type and missing-value information.	Medium	PASS: Displayed statistics match the dataset. FAIL: Statistics are inaccurate or refer to another dataset.	Helps users understand data quality.
FR-2.5	Functional	The system shall allow a user to replace or upload a dataset while preserving correct ownership and metadata associations.	Medium	PASS: Dataset metadata remains associated with the correct owner. FAIL: Ownership is lost or another user's data is exposed.	Supports dataset updates and integrity.
FR-3.1	Functional	The system shall allow users to configure and apply supported data-preprocessing operations before training.	High	PASS: Supported options are accepted and applied or recorded. FAIL: Invalid options are accepted or valid options fail unexpectedly.	Prepares data for ML.
FR-3.2	Functional	The system shall allow a user to select a valid target column from the dataset.	High	PASS: An existing usable column is saved as target. FAIL: A nonexistent or unusable column is accepted.	Defines the prediction target.
FR-3.3	Functional	The system shall allow a user to configure a supported train-test split.	High	PASS: A valid split is accepted and yields usable partitions. FAIL: An invalid split is accepted.	Separates training and evaluation data.
FR-4.1	Functional	The system shall present supported ML algorithms and allow selection of one compatible with the configured task and data.	High	PASS: A supported compatible algorithm is selected. FAIL: Unsupported or incompatible algorithm is accepted.	Enables algorithm selection.
FR-4.2	Functional	The system shall train the selected model using the chosen dataset and experiment configuration when training is started.	High	PASS: Training uses the selected configuration and reports status. FAIL: Incorrect configuration is used or status is absent.	Core evaluator function.
FR-4.3	Functional	The system shall record each training run's status and available configuration details and associate them with the correct user.	High	PASS: Run details are retrievable and correctly associated. FAIL: Details are missing, incorrect, or misassociated.	Supports traceability.
FR-5.1	Functional	The system shall calculate and display evaluation metrics appropriate to the completed run's task type.	High	PASS: Appropriate classification or regression metrics are shown. FAIL: Metrics are missing or inappropriate.	Measures model performance.
FR-5.2	Functional	The system shall display supported evaluation visualizations, such as a confusion matrix where applicable.	Medium	PASS: A compatible visualization reflects the selected run. FAIL: Visualization is incorrect or inapplicable.	Makes results easier to interpret.
FR-5.3	Functional	The system shall compare two or more accessible, compatible runs using comparable metrics and configuration details.	Medium	PASS: Comparable runs show correct metrics. FAIL: Incompatible runs are misleadingly compared.	Supports model selection.
FR-5.4	Functional	The system shall allow users to download supported model artifacts or evaluation reports for runs they may access.	Medium	PASS: The correct available output is downloaded. FAIL: Wrong or unauthorized output is returned.	Allows retention of results.

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FR-6.1	Functional	The system shall display the authenticated user's available run history with recorded status and result details.	Medium	PASS: The user's runs and details are shown accurately. FAIL: Runs are missing, inaccurate, or expose another user's history.	Enables review of prior experiments.
FR-6.2	Functional	The system shall expose documented backend functionality through a defined API/interface for frontend requests.	Medium	PASS: Valid requests return the expected response structure and invalid requests return controlled errors. FAIL: Requests are handled inconsistently.	Defines frontend-backend communication.
FR-6.3	Functional	The system shall provide understandable error feedback for invalid input and failed operations without exposing sensitive internal details.	High	PASS: Actionable feedback is shown without secrets or stack traces. FAIL: Errors are silent or reveal sensitive details.	Improves usability and safe failure handling.
FR-7.1	Functional	The system shall provide authorized Admin users with the user-management functions included in the project scope.	Medium	PASS: Admin can access supported functions and other roles are denied. FAIL: Unauthorized access is allowed.	Supports administration.
FR-7.2	Functional	The system shall provide authorized Admin users with access to dataset records and metadata for supported management.	Medium	PASS: Records are accurate and ownership associations remain intact. FAIL: Unauthorized access or inconsistent ownership occurs.	Supports dataset administration.
FR-7.3	Functional	The system shall provide authorized Admin users with read-only access to available application activity logs.	Low	PASS: Admin can view logs and non-Admins are denied. FAIL: Unauthorized access or application-level log editing is allowed.	Supports accountability.
FR-7.4	Functional	The system shall allow an Instructor to view evaluation results explicitly made available for review, without modifying datasets or runs.	Low	PASS: Authorized results are read-only. FAIL: Unauthorized results are visible or writes succeed.	Supports limited review.
NFR-1	Nonfunctional (Performance)	For supported datasets within the configured size limit, common UI actions and metadata/preview requests shall return within an agreed target under normal load; the target shall be measured and documented during testing.	High	PASS: Response-time measurements are recorded and compared with the agreed project target under normal load. FAIL: The agreed target is not met or no measurements are recorded.	Keeps common user actions responsive.
NFR-2	Nonfunctional (Performance)	The system shall provide visible progress or a clear running status for model-training operations that take longer than a normal request.	Medium	PASS: The user can identify when training is running, completed, or failed. FAIL: A long operation appears unresponsive without status feedback.	Improves transparency during training.
NFR-3	Nonfunctional (Security)	Passwords shall not be stored in plaintext; authentication credentials shall be handled using an appropriate secure password-storage mechanism.	High	PASS: Stored passwords use an appropriate secure representation and are not exposed in responses or logs. FAIL: Plaintext passwords are stored or exposed.	Protects user accounts.
NFR-4	Nonfunctional (Security)	Server-side authorization shall protect user-specific datasets, model runs, and downloadable artifacts; hiding a UI control alone shall not count as access control.	High	PASS: Unauthorized direct backend requests are denied. FAIL: Protected resources can be accessed by bypassing UI restrictions.	Prevents unauthorized access.
NFR-5	Nonfunctional (Privacy)	The system shall isolate each user's uploaded datasets, run history, metrics, and artifacts from other users unless explicit sharing is implemented.	High	PASS: Cross-user requests for restricted resources are denied. FAIL: Another user's datasets, runs, metrics, or artifacts are exposed.	Protects user data and results.
NFR-6	Nonfunctional (Reliability)	Invalid input or a failed training/export operation shall not corrupt previously saved datasets or completed run records.	High	PASS: Failed operations are reported and existing datasets/completed runs remain intact. FAIL: Previously saved data or runs are corrupted.	Maintains data integrity during failures.
NFR-7	Nonfunctional (Reproducibility)	The system shall retain sufficient run configuration, including algorithm and relevant split/preprocessing settings, to explain how a result was produced.	Medium	PASS: A prior run's stored configuration and results can be retrieved and interpreted. FAIL: Results cannot be linked to the configuration used.	Supports interpretation of previous experiments.
NFR-8	Nonfunctional (Usability)	The primary workflow shall be navigable through a desktop web UI with clear labels, validation messages, and status feedback.	Medium	PASS: A tester can complete the primary workflow and understand validation/status feedback. FAIL: Core steps are unclear or feedback is missing.	Makes the application usable for its intended workflow.
NFR-9	Nonfunctional (Compatibility)	The application shall support the agreed current versions of major desktop browsers; the exact test matrix shall be recorded for the release.	Medium	PASS: Core workflows operate in each browser/version in the agreed test matrix. FAIL: A core workflow is unusable in a supported browser.	Supports browser-based access.
NFR-10	Nonfunctional (Maintainability)	The implementation shall separate UI, API/business logic, data persistence, and model-training concerns into maintainable modules.	Medium	PASS: The major responsibilities are separated into identifiable modules. FAIL: Routine changes require modifying unrelated parts of the application.	Supports maintainable development.
NFR-11	Nonfunctional (Scalability)	The system shall enforce configurable upload/resource limits and handle datasets within the documented project capacity without uncontrolled resource use.	Medium	PASS: Configured limits are enforced and over-limit inputs receive controlled feedback. FAIL: Limits are ignored or resource use becomes uncontrolled.	Helps control resource consumption.
NFR-12	Nonfunctional (Data Integrity)	Dataset metadata, run configuration, metrics, and artifact references shall remain consistently associated with the correct user and run.	High	PASS: Metadata, results, and artifacts refer to the correct user and run. FAIL: References are missing, inconsistent, or misassociated.	Maintains trustworthy experiment records.

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NFR-13	Nonfunctional (Error Handling)	Errors shown to users shall be understandable and shall not expose passwords, secret configuration, stack traces, or other sensitive internal details.	High	PASS: Error messages are actionable and omit sensitive internal information. FAIL: Errors expose secrets, stack traces, or other sensitive details.	Supports safe troubleshooting.
NFR-14	Nonfunctional (Portability)	The application shall be deployable in the documented project environment using its setup instructions and declared dependencies.	Medium	PASS: A clean environment can be set up using the documented instructions and declared dependencies, or gaps are recorded. FAIL: Setup instructions or dependency declarations prevent reproducible setup.	Supports reliable setup and deployment.